Kasetsart University Siracha Campus

PYTHON PROGRAMMING PROJECT ASSIGNMENT

AI Trash Detection and Monitoring System

By Shine Lin Htet
Student ID - 6730161035

List of Content

1. Introduction
2. Problem and Solution
3. Target Audience
4. Key Features
5. Code explanation (Flow Chart)
6. Technology Stack
7. Project Demostration
(GUI features, Data log, Email Alert, InfluxDB)
8. Further Improvement

Introduction

In many urban areas, including river roads and public spaces, large amounts of trash are irresponsibly disposed of every day. These waste items not only pollute the environment but also harm wildlife and affect community health.

Challenges with Human Capabilities

Manual monitoring of such areas is nearly impossible — **human eyes** cannot watch every corner at all times.

What if we create a computer vision AI which can monitor the trash in real time?

This is where Artificial Intelligence (AI) can play a powerful role.

Solution

Designed this Project to fulfill the human need

This project introduces a real-time trash detection system powered by AI.
Using computer vision, the system can

- Detect trash instantly through a live camera feed.
- Track types and quantities of detected waste.
- Log and visualize data in real time.
- Send alerts when trash activity is high.

.

Target Audiences and Applications

Environmental Monitoring

- Monitor trash levels in real-time to optimize garbage collection schedules.
- Track types and frequency of trash for research and pollution trend analysis.

Municipal Authorities

Environmental Agencies

Smart City Developers

Key Features

🎥 Real-Time Trash Detection
Detects and labels trash items live using webcam and Roboflow (YOLO).

📊 Live Data Visualization
- Pie chart (trash type distribution)
- Line graph (trash detection trend per minute)

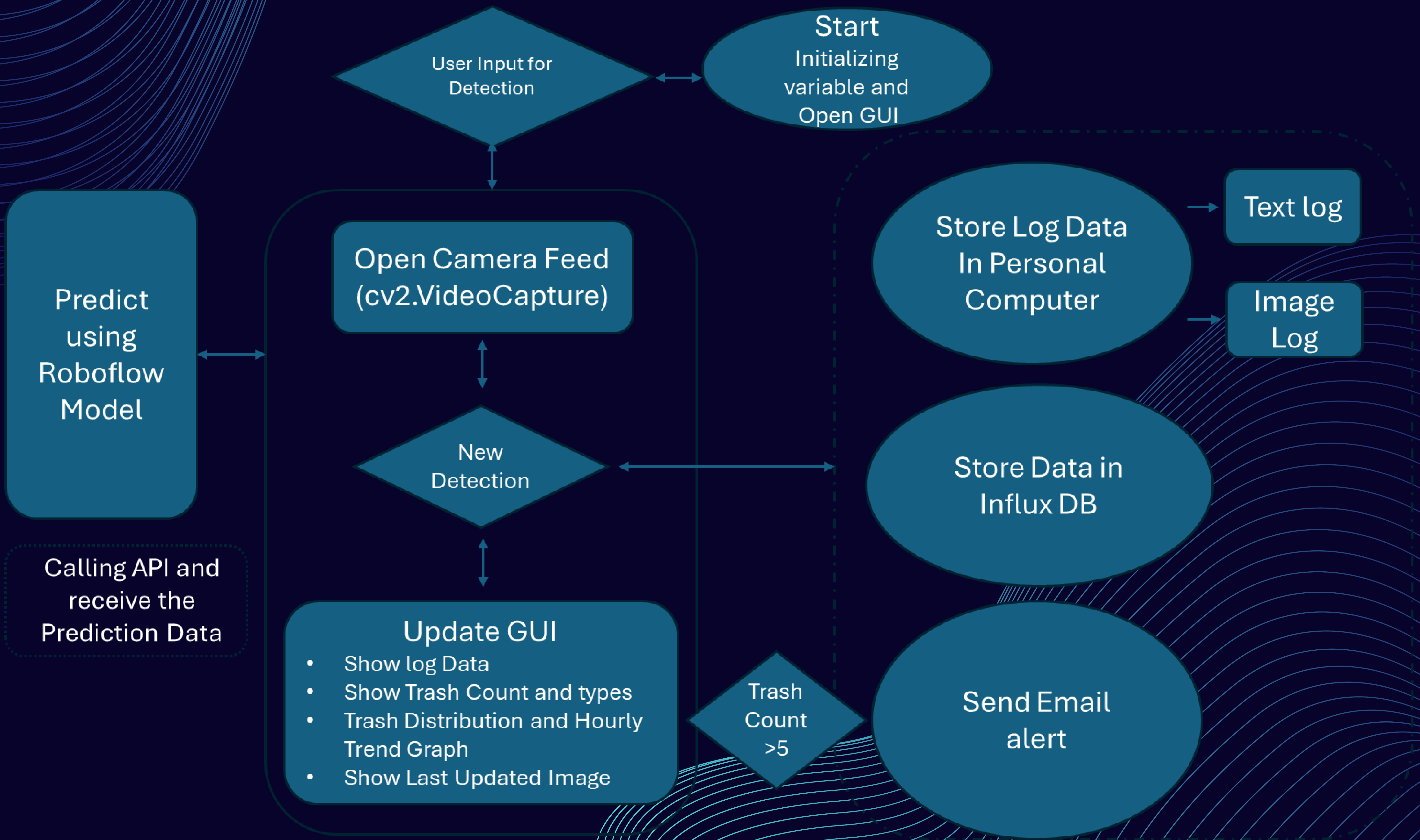
📤 InfluxDB Integration
Logs detection data (type, confidence, timestamp) for historical analysis.

📧 Email Alerts
Sends automatic email when high trash activity is detected in a short time.

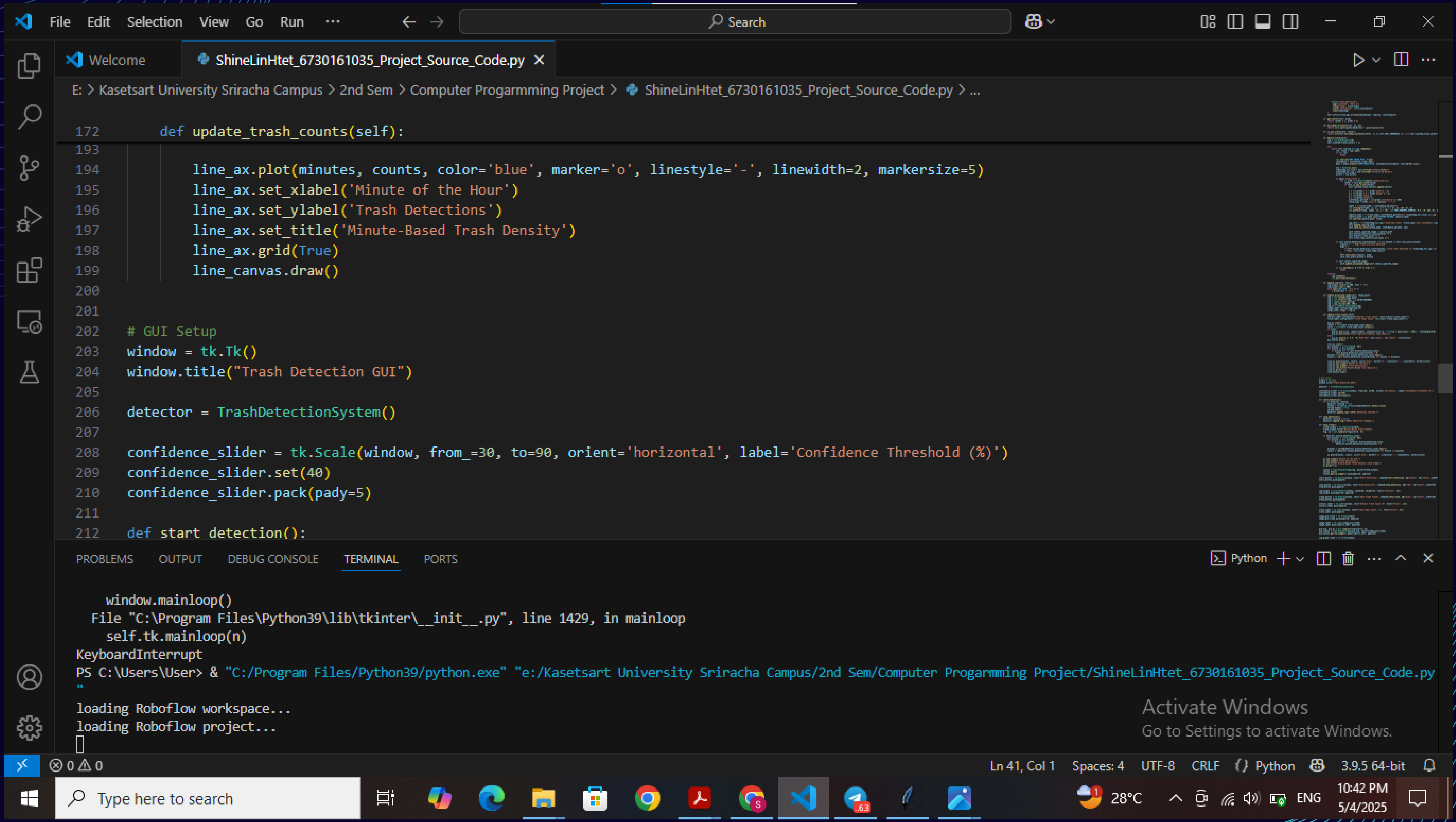
🗃️ Image Capture & Logging
Saves labeled images and logs detection results to a text file.

🖱️ User-Friendly GUI (Tkinter)
Start/Stop detection, live logs, image preview, graph trend, and confidence adjustment.

Code Explanation(Flow Chart)



Python Code



File Edit Selection View Go Run ···
Search
Welcome
ShineLinHtet_6730161035_Project_Source_Code.py
E: > Kasetsart University Sriracha Campus > 2nd Sem > Computer Progarmming Project > ShineLinHtet_6730161035_Project_Source_Code.py > ...
172 def update_trash_counts(self):
193
194 line_ax.plot(minutes, counts, color='blue', marker='o', linestyle='-', linewidth=2, markersize=5)
195 line_ax.set_xlabel('Minute of the Hour')
196 line_ax.set_ylabel('Trash Detections')
197 line_ax.set_title('Minute-Based Trash Density')
198 line_ax.grid(True)
199 line_canvas.draw()
202 # GUI Setup
203 window = tk.Tk()
204 window.title("Trash Detection GUI")
206 detector = TrashDetectionSystem()
208 confidence_slider = tk.Scale(window, from_=30, to=90, orient='horizontal', label='Confidence Threshold (%)')
209 confidence_slider.set(40)
210 confidence_slider.pack(pady=5)
212 def start_detection():
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
window.mainloop()
File "C:\Program Files\Python39\lib\tkinter__init__.py", line 1429, in mainloop
self.tk.mainloop(n)
KeyboardInterrupt
PS C:\Users\User> & "C:/Program Files/Python39/python.exe" "e:/Kasetsart University Sriracha Campus/2nd Sem/Computer Progarmming Project/ShineLinHtet_6730161035_Project_Source_Code.py"
loading Roboflow workspace...
loading Roboflow project...
Activate Windows
Go to Settings to activate Windows.
Ln 41, Col 1 Spaces: 4 UTF-8 CRLF Python 3.9.5 64-bit
Type here to search
28°C ENG 10:42 PM 5/4/2025

Technology Stack

👨‍💻 Programming Language
Python – Core development language for detection logic, GUI, and data handling.

🧠 AI & Computer Vision
Roboflow – For object detection model and prediction API.
OpenCV – For image processing, webcam access, and drawing detections.

🖼️ GUI Development
Tkinter – For building the interactive user interface.
Matplotlib – For real-time graph visualizations (pie chart and line graph).
PIL (Pillow) – For displaying images in the GUI.

📦 Data Storage & Analytics
InfluxDB – Time-series database for storing trash detection logs.

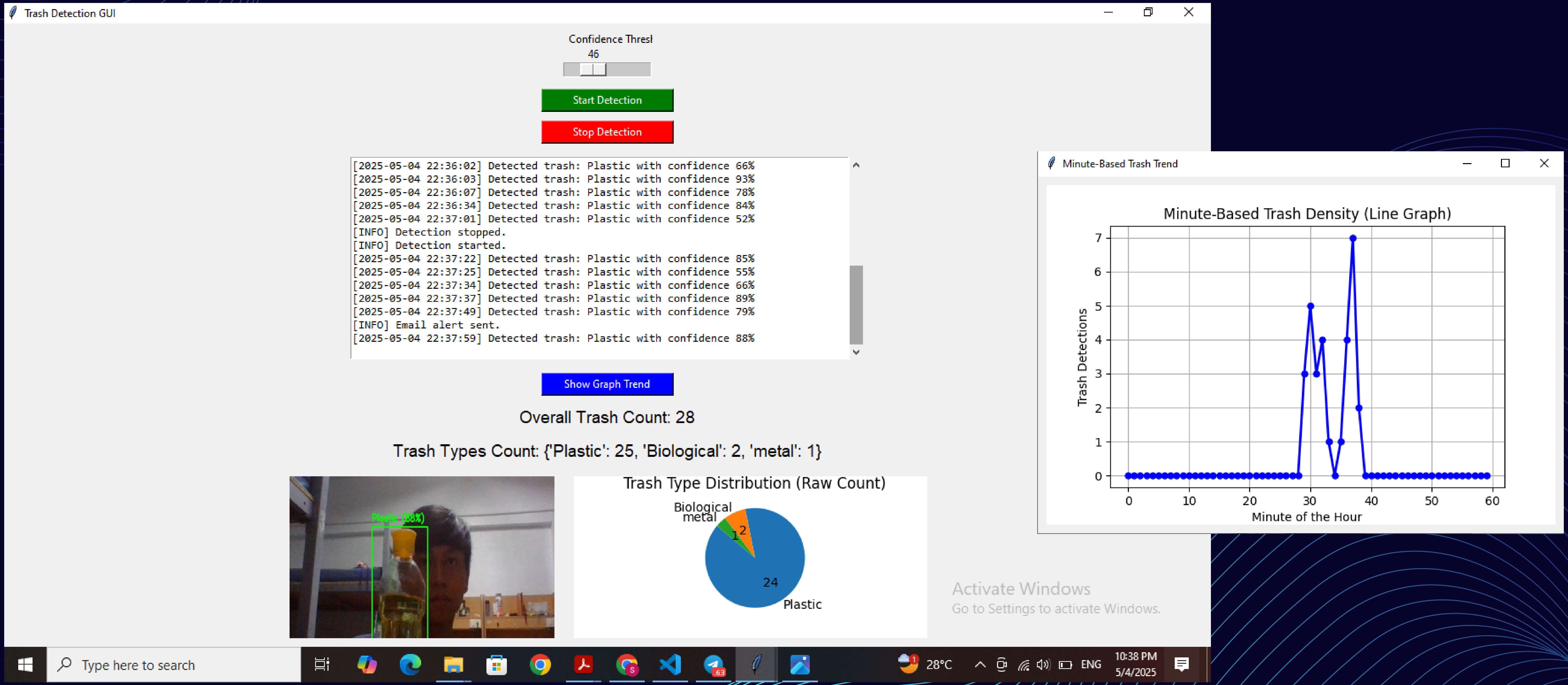
📧 Email Notification
smtplib + email.mime – For sending alert emails when trash activity spikes.

⚙️ Concurrency
threading – To run detection in the background without freezing the GUI.

🧱 Support Libraries
collections – For counters and structured data.
datetime, os, math – For timestamps, file handling, and geometric calculations.

Project Demostration

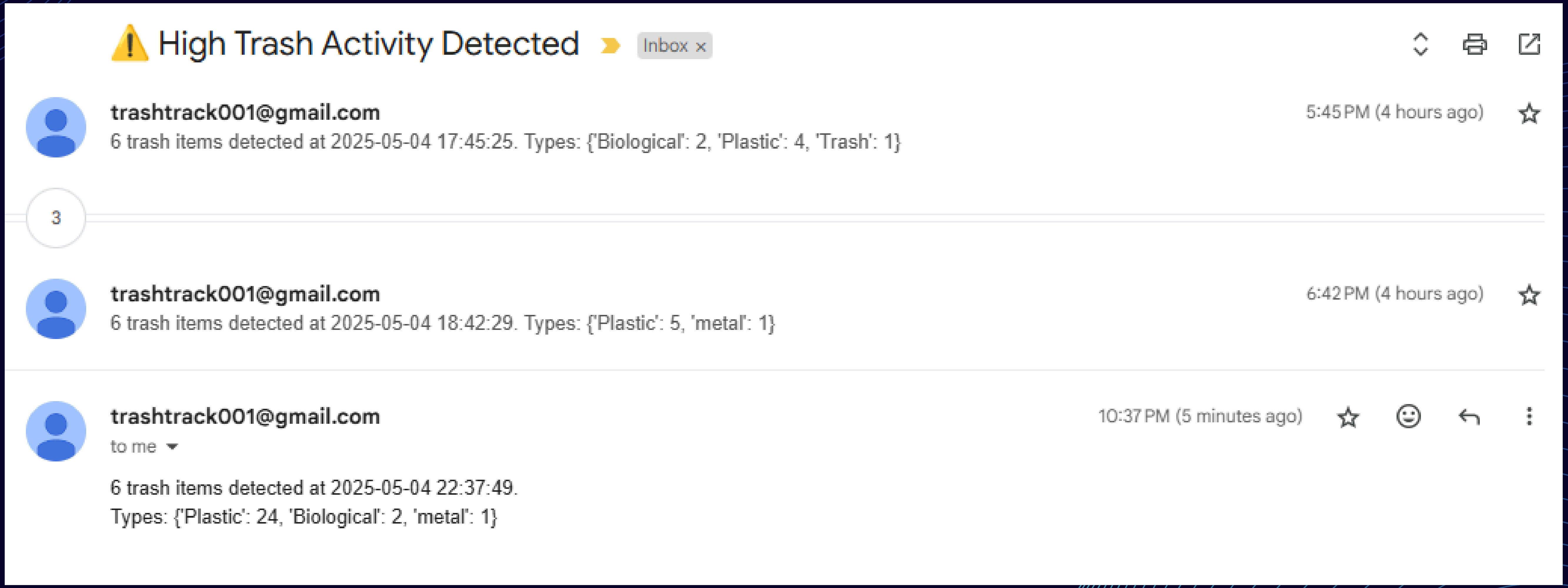
Interactive GUI Dashboard



Project Demostration

Email Alert System

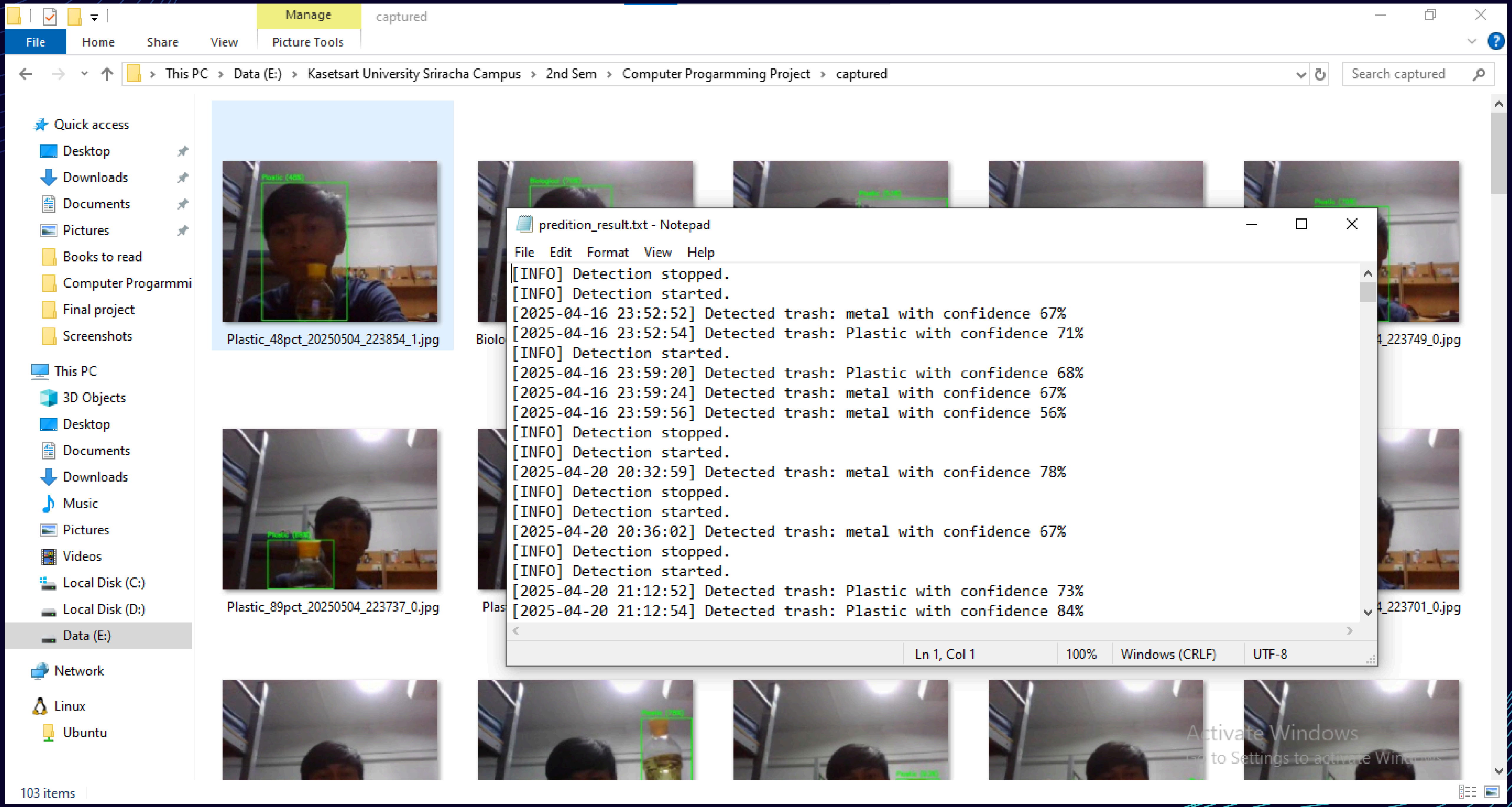
- When Trash Volume is High, system will send the email notification automatically.



Project Demostration

Basically, data log are stored as two types in local computer,

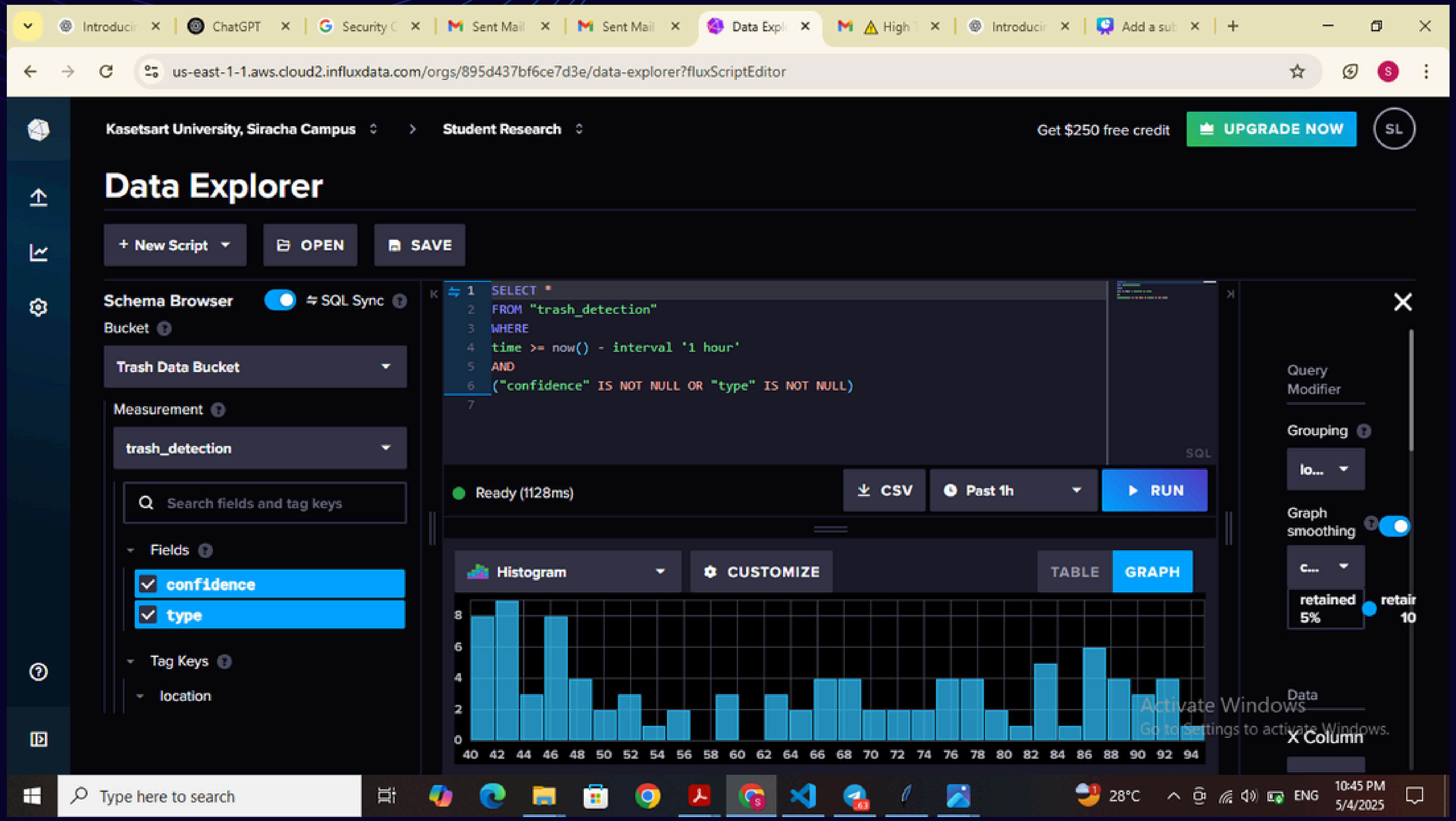
- Image Log data (jpg file)
- Text Log Data (txt file)

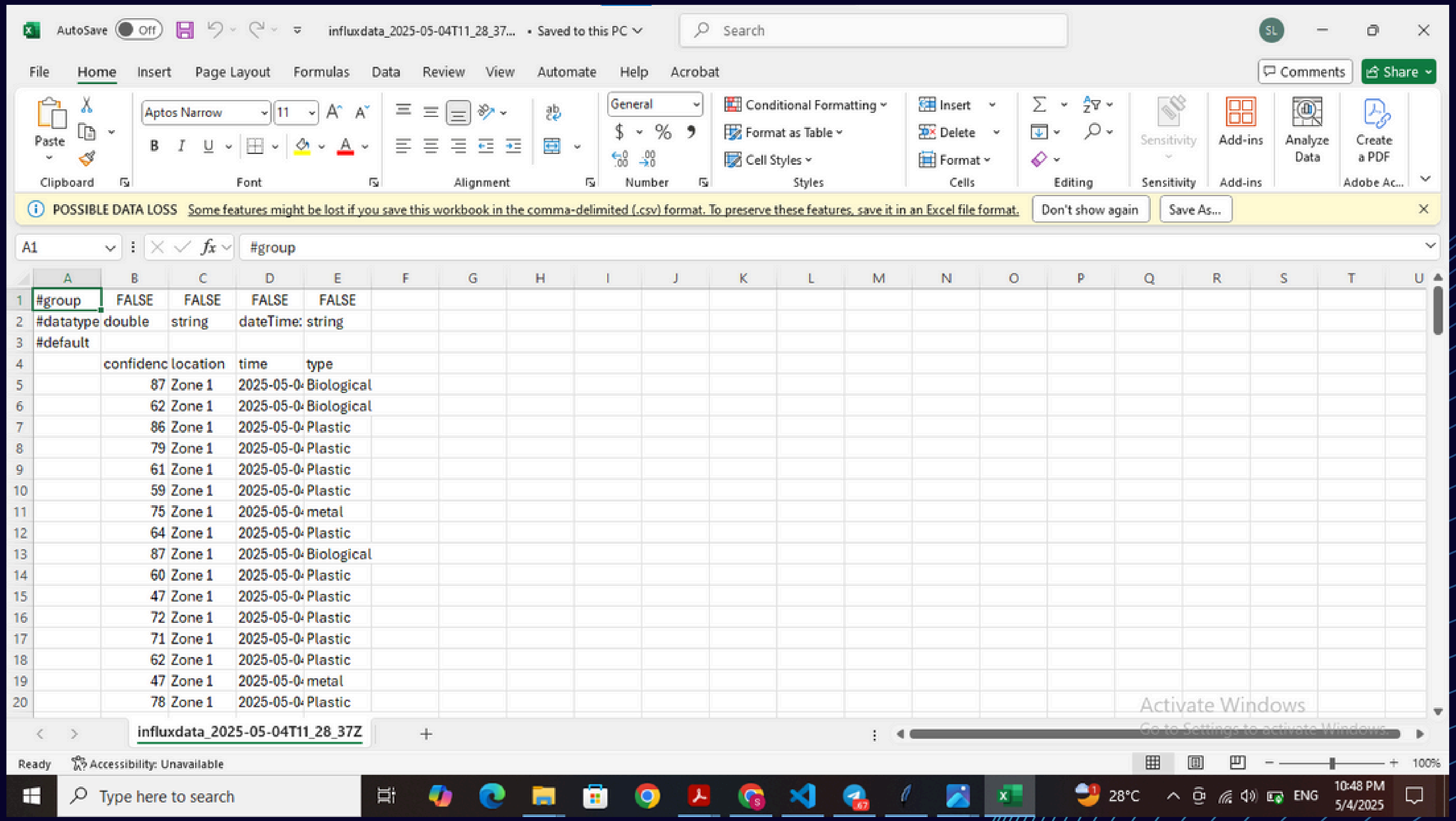


Project Demostration

Real Time Could Data storage (InfluxDB)

Send the Logs detection data (type, confidence, timestamp) to InfluxDB for historical analysis.



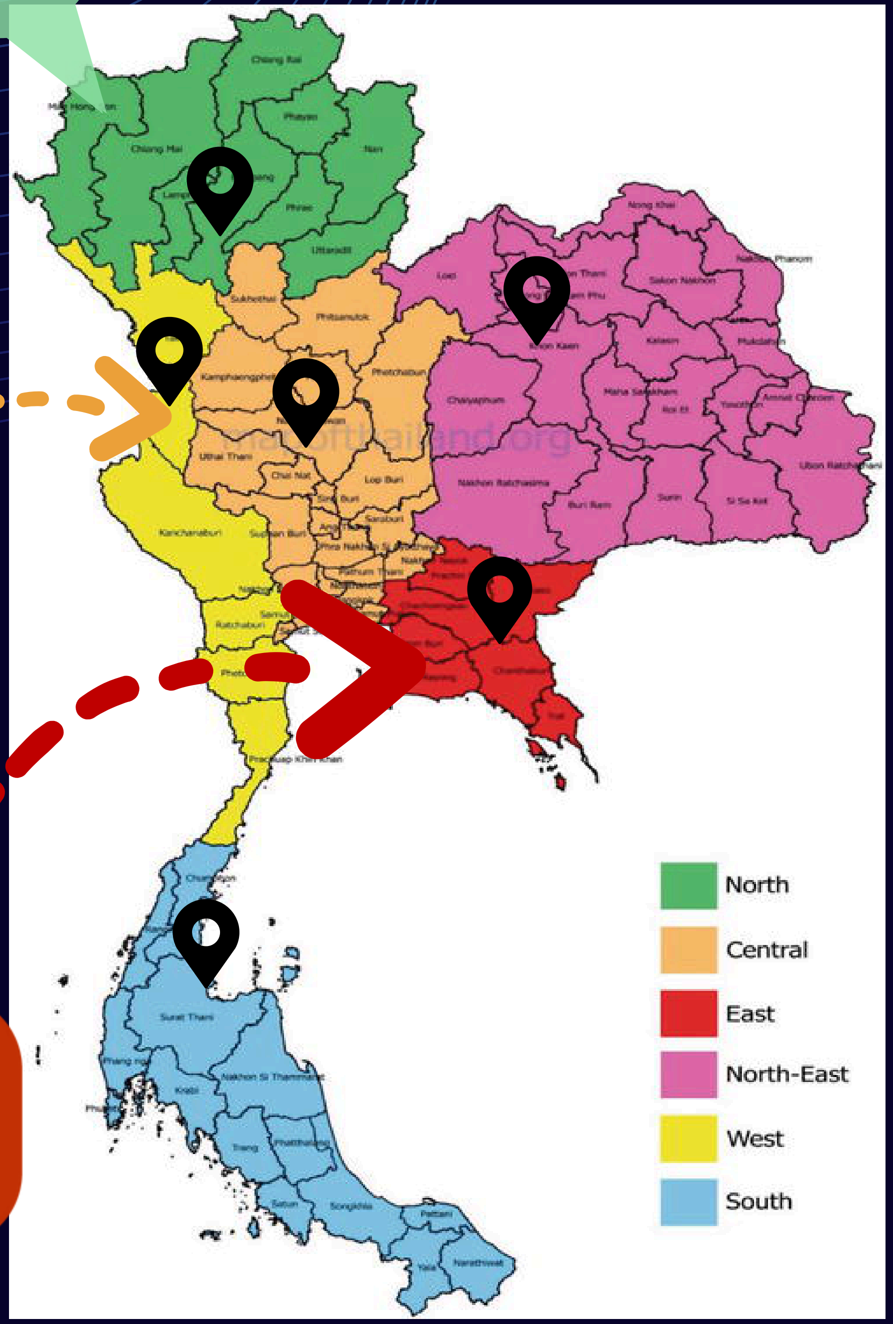


Further Improvement

Zone 2
LOW Trash volume

Zone 3
Medium Trash volume

Zone 1
High Trash volume

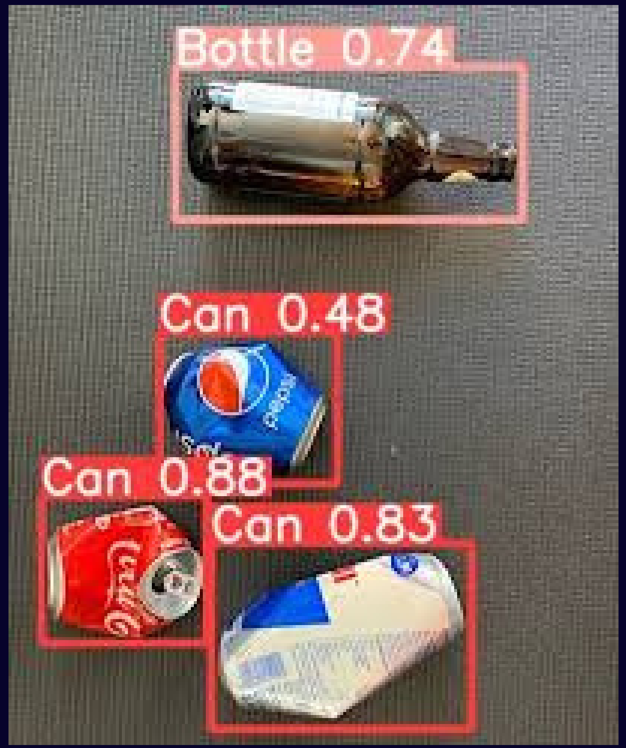


- **Multi-CCTV camera integration with geographical data (GPS)**

Location	Trash volume	Action
Zone1	HIGH	HIGH priority
Zone2	LOW	-
Zone3	MEDIUM	LOW priority
Zone4	LOW	-

Integration of GPS system to monitor multiple geographical zones simultaneously using multiple CCTV camera.

- **AI Model Enhancement**



Train a custom Roboflow model with more trash types and real-world images for better accuracy.

Thank You for your Time